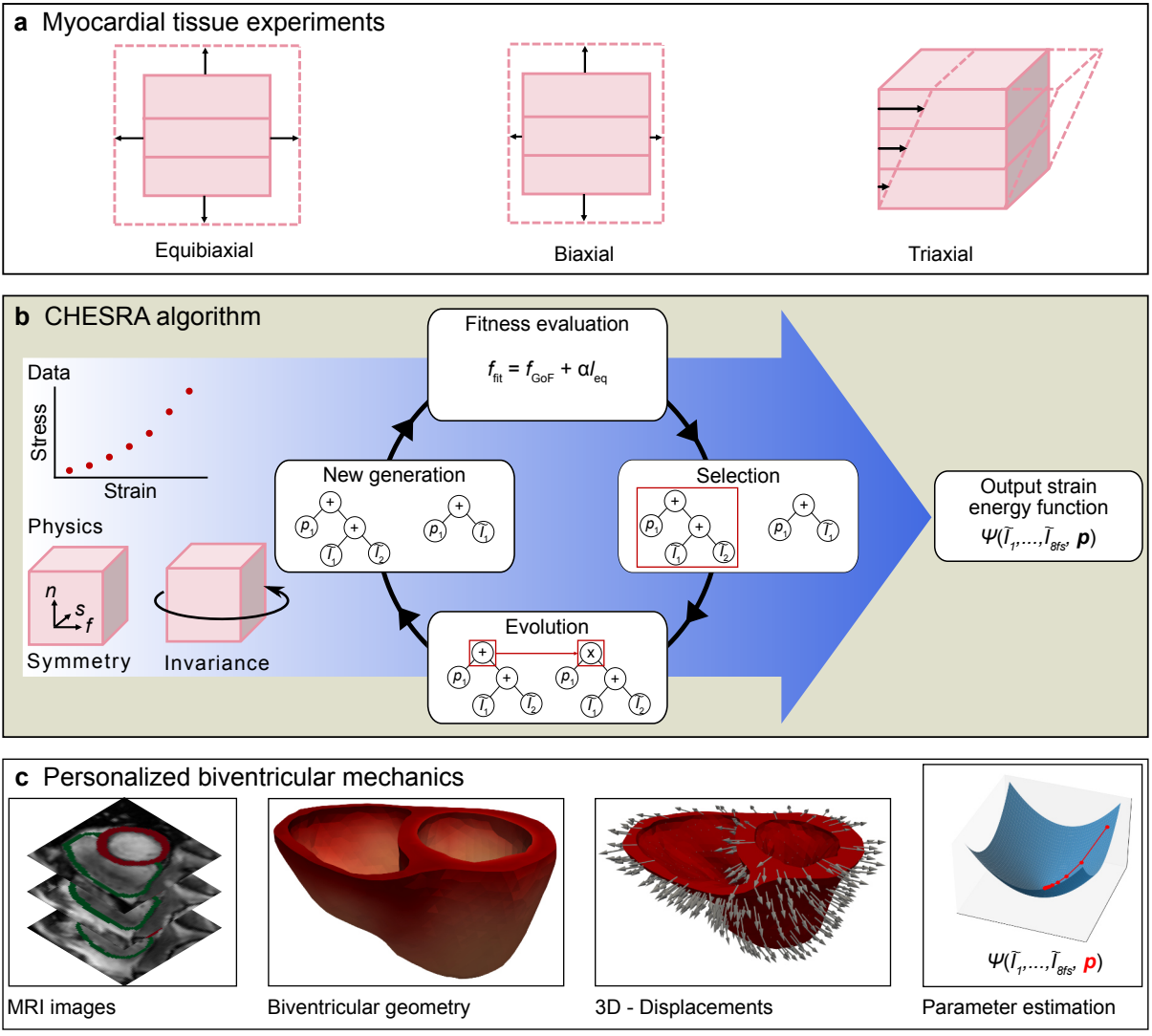
a Myocardial tissue experiments
Equibiaxial
Biaxial
Triaxial
b CHESRA algorithm
Data
Stress
Strain
Physics
Symmetry
Invariance
Fitness evaluation
$f_{fit} = f_{GoF} + \alpha I_{eq}$
New generation
Selection
Evolution
Output strain energy function
$\Psi(\bar{I}_1,\ldots,\bar{I}_{8fs}, \boldsymbol{p})$
c Personalized biventricular mechanics
$\Psi(\bar{I}_1,\ldots,\bar{I}_{8fs}, \boldsymbol{p})$
MRI images
Biventricular geometry
3D - Displacements
Parameter estimation